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Dimensionality crossover to a two-dimensional vestigial nematic state from a three-dimensional antiferromagnet in a honeycomb van der Waals magnet

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The effects of fluctuations and disorder, which are substantially enhanced in reduced dimensionalities, can play a crucial role in producing non-trivial phases of matter such as vestigial orders characterized by a composite order parameter. However, fluctuation-driven magnetic phases in low dimensions have remained relatively unexplored. Here we demonstrate a phase transition from the zigzag antiferromagnetic order in the three-dimensional bulk to a Z_3 vestigial Potts nematicity in two-dimensional few-layer samples of van der Waals magnet $NiPS_3$. Our spin relaxometry and optical spectroscopy measurements reveal that the spin fluctuations are enhanced over the gigahertz to terahertz range as the layer number of $NiPS_3$ reduces. Monte Carlo simulations corroborate the experimental finding of threefold rotational symmetry breaking but show that the translational symmetry is restored in thin layers of $NiPS_3$. Therefore, our results show that strong quantum fluctuations can stabilize an unconventional magnetic phase after destroying a more conventional one.

A vestigial order is due to the partial melting of the primary order (η) of a spontaneous symmetry-breaking phase[1,2]. In contrast to conventional short-range orders that do not break symmetries of their hosting crystalline lattices, a vestigial order corresponds to a composite order ($\eta^\dagger\tau\eta$, a quadratic form of η with a matrix τ) and breaks a subset of symmetries broken by the primary order[1,2]. Fluctuations and disorders are the typical causes for destroying primary orders and, potentially, introducing vestigial orders. As illustrated in Fig. 1a, an increase in temperature leads to thermal fluctuations, and the reduced dimensionality can independently contribute to enhanced fluctuations. Between the long-range primary ordered region I ($\langle\eta\rangle \neq 0$ and $\langle\eta^\dagger\tau\eta\rangle \neq 0$) and the completely disordered region III ($\langle\eta\rangle = 0$ and $\langle\eta^\dagger\tau\eta\rangle = 0$), there could exist region II for the vestigial ordered case[1,2] ($\langle\eta\rangle = 0$ but $\langle\eta^\dagger\tau\eta\rangle \neq 0$). The impact of introducing vestigial order has been profound, as it leads to the intertwined nature of proximate phases within the rich phase diagrams of many quantum material systems

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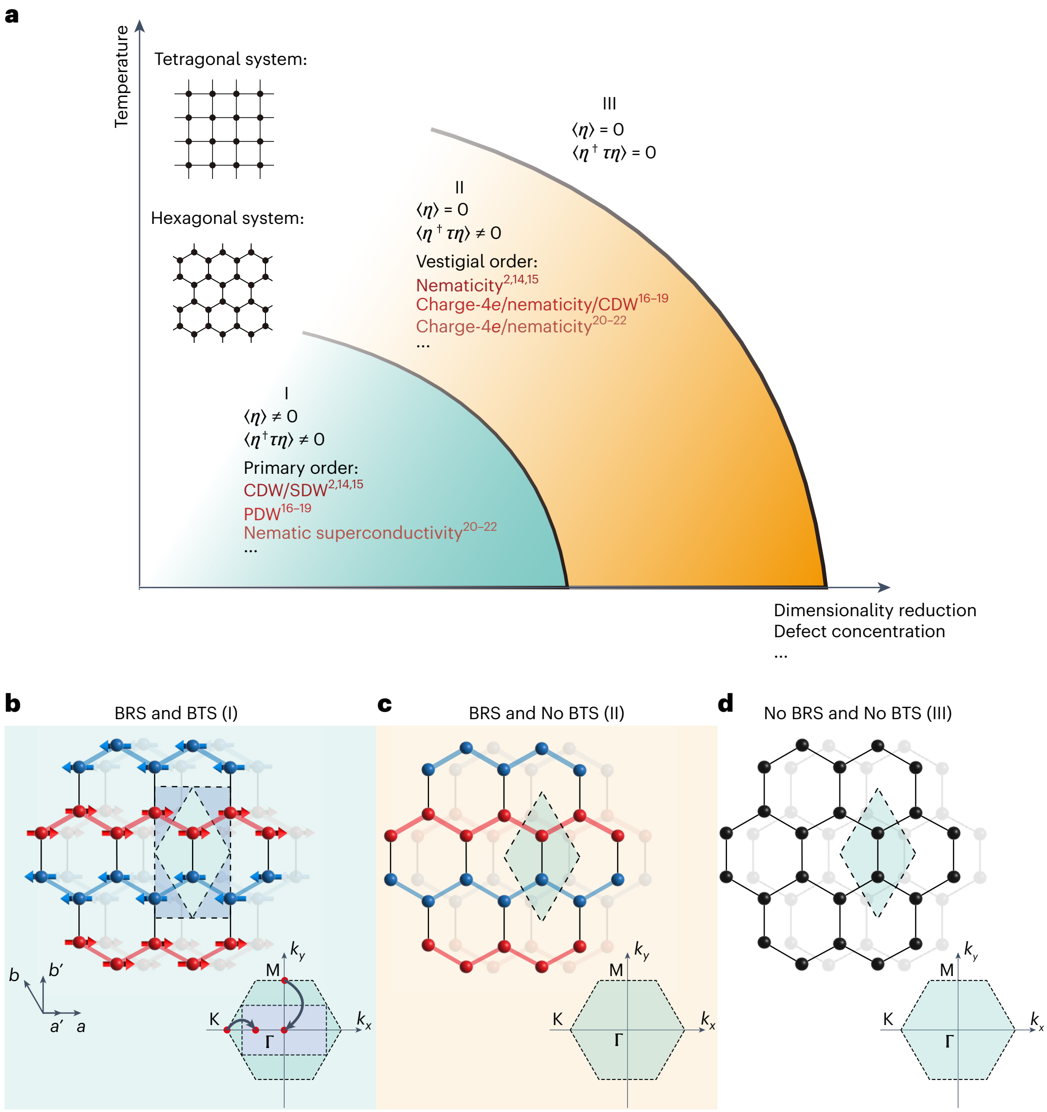

Fig. 1 | Schematic phase diagrams of vestigial order and magnetic states in $NiPS_3$. **a**, Schematic phase diagram showing the primary order (η) and the vestigial order ($\eta^\dagger\tau\eta$) as functions of temperature (vertical axis) and other independent factors, such as enhanced fluctuations at reduced dimensionality or with disorder (horizontal axis). Three distinct regions are shown: (I) long-range primary order ($\langle\eta\rangle \neq 0$ and $\langle\eta^\dagger\tau\eta\rangle \neq 0$, shaded with blue), (II) long-range vestigial order with short-range primary order ($\langle\eta\rangle = 0$ but $\langle\eta^\dagger\tau\eta\rangle \neq 0$, shaded with orange) and (III) disorder ($\langle\eta\rangle = 0$ and $\langle\eta^\dagger\tau\eta\rangle = 0$). Examples of primary order versus vestigial order are listed in the phase diagram, including charge or spin density wave versus nematicity[2,14,15], pair density wave versus charge-4*e* superconductivity or nematicity or charge density wave[16–19], superconductivity versus charge-4*e* superconductivity or nematicity[20–22], and so on. **b**, Schematic of the long-range zigzag AFM order in 3D $NiPS_3$ with both BRS and BTS, corresponding to region I in **a**. The Ni^{2+} spins are aligned along the zigzag chain. The green parallelogram represents the in-plane unit cell of the crystal structure without the AFM order, and the blue rectangle shows the doubled in-plane unit cell for the zigzag AFM state. Inset, BZ folding in momentum space. The M point at the BZ boundary without the AFM order folds to the Γ point of the BZ centre due to the zigzag AFM order. **c**, Schematic of the spin-induced nematic state in 2D $NiPS_3$ with BRS but without BTS, corresponding to region II in **a**. Inset, corresponding BZ without folding due to the lack of BTS. **d**, Schematic of the spin disordered state without BRS and BTS in $NiPS_3$, corresponding to region III in **a**. Inset, corresponding BZ. In **b**–**d**, only the Ni atoms are shown for simplicity. CDW, charge density wave; PDW, pair density wave; SDW, spin density wave.

including Cu-based high-T_c superconductors[1,3–7], Fe-based superconductors[8–11], charge density wave systems[12] and others[13]. Theoretical examples of primary versus vestigial orders are common in many systems: charge or spin density wave versus nematicity[2,14,15], pair density wave versus charge-4*e* superconductivity or nematicity or charge density wave[16–19], superconductivity versus charge-4*e* superconductivity or nematicity[20–22], and even the anyon condensation from a $Z_2 \times Z_2$ topological order quantum spin liquid versus Z_2 topological order quantum spin liquid[23], and so on.

Experimental realizations of vestigial order have primarily focused on the nematicity that breaks the rotational symmetry but preserves the translational symmetries of underlying crystal lattices[2,15]. This symmetry requirement allows tetragonal and hexagonal systems to support the emergence of nematicity. Indeed, in tetragonal systems such as cuprates[6,7] and pnictides[9,15], Z_2 Ising nematicity can develop from the charge and spin fluctuations, which breaks the tetragonal fourfold symmetry into orthorhombic twofold rotational symmetry while retaining the translation symmetries. In contrast, in hexagonal systems, Z_3 Potts nematicity is anticipated. This breaks the hexagonal sixfold (or threefold) rotational symmetry and preserves the translational symmetries[24–27]. Very recently, in systems such as bulk $Fe_{1/3}NbS_2$ (ref. 28) and bulk $FePS_3$ (ref. 29), the characteristics of Potts nematicity have been seen in their long-range ordered magnetic phases, that is, region I in Fig. 1a. However, the intrinsic Potts-nematic order without the primary order, that is, region II in Fig. 1a, has been much less explored. Z_2 Ising nematicity and Z_3 Potts nematicity are also expected to show distinct phase transitions upon the application of an external strain[30].

Reducing dimensionality is one promising but less explored route for enhancing fluctuations. In fact, pioneering theoretical studies on nematicity have often used two-dimensional (2D) models[2]. In this regard, we chose a 2D honeycomb magnetic system with the XY-type spin anisotropy, $NiPS_3$ (refs. 31–35), to harvest stronger spin fluctuations for realizing Z_3 Potts nematicity. This study is distinct from mainstream research on 2D magnetism, which aims to maintain long-range magnetic order against fluctuations[36–40]. Monolayer $NiPS_3$ has the trigonal crystallographic point group D_{3d} with an out-of-plane threefold rotational axis. Bulk and few-layer $NiPS_3$ obey the monoclinic structural point group C_{2h} due to the lateral shift between adjacent layers along the a axis. Bulk $NiPS_3$ undergoes an antiferromagnetic (AFM) transition at $T_{N,3D}$ = 155 K. In the long-range ordered AFM state, the Ni^{2+} spins align along the a axis ferromagnetically within the zigzag chains and are coupled AFM between neighbouring chains[41–44] (Fig. 1b). Such a long-range zigzag AFM order in $NiPS_3$ spontaneously breaks the threefold rotational symmetry of individual layers, and at the same time, breaks the translational symmetry with a wavevector $\mathbf{Q} = \mathbf{k}_M$, where $\mathbf{k}_M$ is the momentum at the M point in the Brillouin zone (BZ; Fig. 1b, inset). Three degenerate zigzag AFM states are expected, with three $\mathbf{Q}$ rotated from each other by 120°, that is, $\mathbf{Q}_j$ (j = 1, 2 or 3). Such a broken rotational symmetry (BRS) and broken translational symmetry (BTS) AFM phase in bulk $NiPS_3$ has a non-zero primary order parameter $\eta_j(\mathbf{r}) = (\mathbf{S}(\mathbf{r}) - \mathbf{S}(\mathbf{r}+\mathbf{e}_j))\,e^{i\mathbf{Q}_j\cdot\mathbf{r}}$ with $\mathbf{e}_j$ being a vector parallel to $\mathbf{Q}_j$ between neighbouring spin sites, and $\mathbf{S}(\mathbf{r})$ being the spin at the site $\mathbf{r}$. This phase also supports a finite secondary order parameter $\eta^\dagger\tau\eta = \left(\frac{\sqrt{3}}{2}(\eta_3^2-\eta_2^2), \frac{1}{2}\left(2\eta_1^2-\eta_2^2-\eta_3^2\right)\right)$, which belongs to region I in Fig. 1a. Due to the weak anisotropy in the XY model, strong spin fluctuations in 2D and ferromagnetic-AFM exchange competitions, few-layer $NiPS_3$ is expected to realize the scenario in region II, where the nematic phase has BRS but no BTS (Fig. 1c), before going into the completely spin disordered, paramagnetic phase of region III (Fig. 1d).

We start by verifying the enhancement of spin fluctuations in thinner $NiPS_3$ samples, using both the nitrogen-vacancy (NV) spin relaxometry and the quasi-elastic scattering (QES) in Raman spectroscopy that cover distinct frequency regimes from gigahertz to terahertz. NV spin relaxometry utilizes the excellent magnetic field sensitivity of NV spin centres to probe the fluctuating magnetic fields at the NV electron spin resonance frequencies (gigahertz) generated by proximal magnetic materials[45–47]. The measured NV spin relaxation rate is proportional to the spectral density of the magnetic field noise, which scales with the spin fluctuation strength[46]. Here, we spatially visualize the spin fluctuations in exfoliated $NiPS_3$ flakes using the wide-field imaging mode of NV spin relaxometry[48]. A representative NV spin relaxation rate map is shown in the inset of Fig. 2a for a four-layer (4L) and a 10-nm-thick $NiPS_3$ flake at 170 K. Figure 2a shows the temperature dependence of the area-averaged NV spin relaxation rate for the 4L and 10 nm $NiPS_3$ flakes. There was a decreasing trend for spin fluctuations at lower temperatures for both samples. To account for the thickness difference of the spin fluctuations, we extracted the magnetic susceptibility χ of 4L and 10 nm $NiPS_3$ samples from the NV spin relaxation rate (Supplementary Information Note 1). Figure 2b plots χ, which is proportional to the spin fluctuations[49], as a function of temperature for the 4L and 10 nm $NiPS_3$ samples. It is evident that χ for 4L $NiPS_3$ was clearly enhanced in comparison with that for 10 nm $NiPS_3$ below ~150 K, whereas it was only slightly larger for the 4L sample than for the 10 nm sample above ~150 K.

Complementarily, Raman QES in $NiPS_3$ arises from spin fluctuations[50] and manifests as a slow delay profile in the Raman spectra for

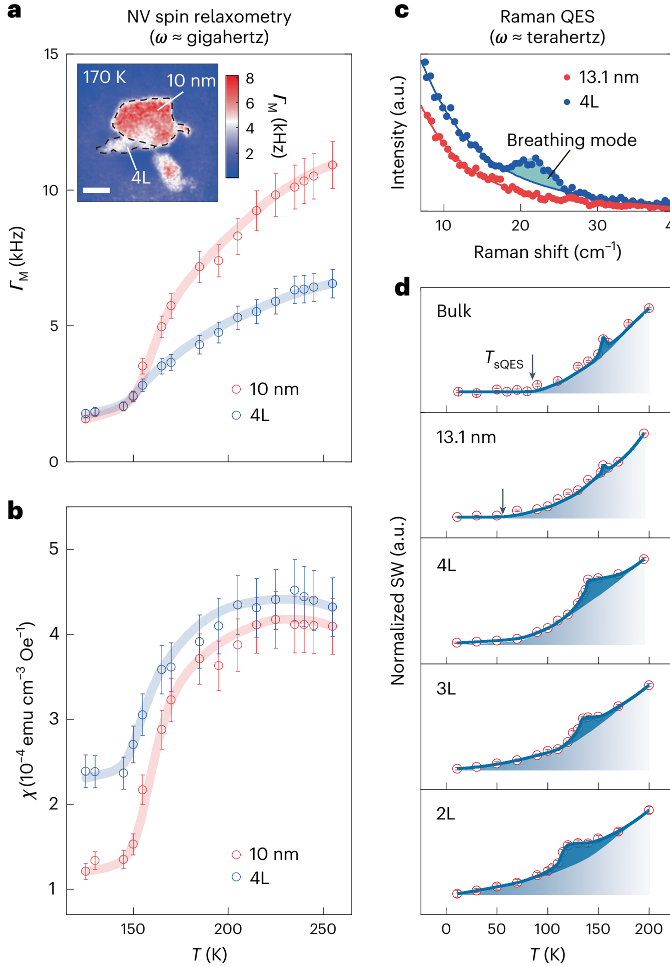

Fig. 2 | Thickness-dependent spin fluctuations in few-layer $NiPS_3$ measured by NV relaxometry and Raman spectroscopy. a, Temperature dependence of NV spin relaxation rate Γ_M measured at NV centres underneath 10 nm and 4L $NiPS_3$. Data points are presented as $\Gamma_M \pm \delta\Gamma_M$, where Γ_M and $\delta\Gamma_M$ denote the optimal fitting parameter and one standard error, respectively. Inset, NV spin relaxation map for the few-layer $NiPS_3$ samples at 170 K. The black dashed lines outline the boundaries of the 10 nm and 4L $NiPS_3$. Scale bar, 4 µm. **b**, Fitted magnetic susceptibility χ of 10 nm and 4L $NiPS_3$ as a function of temperature. Data points are presented as $\chi \pm \delta\chi$, where χ and $\delta\chi$ denote the optimal fitting parameter and one standard error, respectively. **c**, Normalized QES Raman spectra measured with λ = 532 nm for 13.1 nm and 4L $NiPS_3$ at 170 K. The shaded area denotes the breathing mode in 4L $NiPS_3$, which was excluded when calculating the QES spectral integral in **d**. **d**, Temperature-dependent integrated QES intensity over 8–40 cm^{-1} for $NiPS_3$ of different thicknesses. The dark blue shading highlights the peak or bump near the nematic transition. The light blue gradient shading highlights the temperature-dependent evolution of the QES. The arrows indicate the temperatures at which QES vanishes for the bulk and 13.1 nm samples. Data points are presented as SW ± δSW, where SW and δSW denote the optimal fitting parameter and one standard error, respectively. a.u., arbitrary units; emu, electromagnetic unit.

up to ~40 cm^{-1} (of the order of terahertz). Two typical Raman spectra of QES for a 4L and a 13.1 nm $NiPS_3$ flake taken at 170 K are shown in Fig. 2c. Note that both spectra were normalized to their corresponding thicknesses to allow a fair comparison of spin fluctuations between them. It is evident that the spin fluctuations were stronger in 4L than in 13.1 nm $NiPS_3$ over a spectral range up to ~40 cm^{-1}, despite the overall decay at higher frequencies for both samples. Moreover, we can integrate the QES spectral weight (SW_{QES}) over 8–40 cm^{-1}, with that from the phonon breathing mode excluded, to quantify the spin fluctuations and track its temperature dependence (see Supplementary Information Note 2 for

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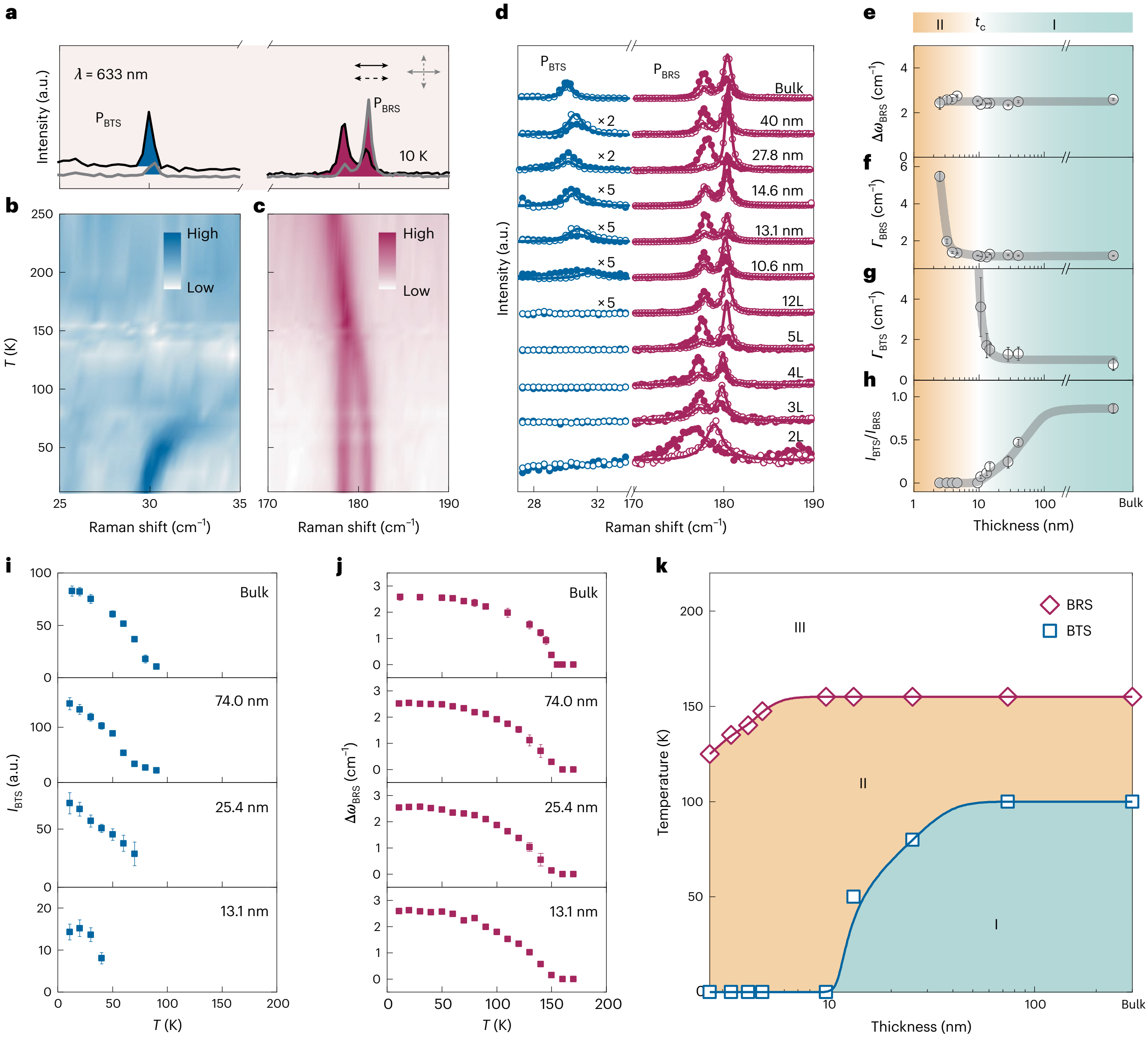

Fig. 3 | Thickness dependence of BTS and BRS signatures in Raman spectroscopy. a, Linearly polarized Raman spectra measured with $\lambda = 633$ nm in both parallel (black curves) and crossed (grey curves) channels at 10 K on bulk $NiPS_3$. The blue and red areas denote the P_{BTS} and P_{BRS} modes, respectively. **b**, Temperature-dependent Raman spectra for the P_{BTS} mode in the parallel channel of bulk $NiPS_3$. To give a clearer presentation of the temperature dependence of the P_{BTS} mode, we subtracted the QES signal from the raw data for temperatures above 90 K. **c**, Temperature-dependent Raman spectra of the P_{BRS} mode in the parallel channel of bulk $NiPS_3$. **d**, Layer-dependent Raman spectra in both parallel (filled circles) and crossed (empty circles) channels at $T = 10$ K. The dark (light) solid lines are the curves fitted using the Lorentz function for the parallel (crossed) channel. **e**, Thickness dependence of fitted frequency separation of P_{BRS}, $\Delta\omega_{BRS} = \omega_{BRS-h} - \omega_{BRS-l}$, at $T = 10$ K. Here, ω_{BRS-h} was fitted for the cross channel and ω_{BRS-l} for the parallel channel. **f**, Thickness dependence of the fitted linewidth Γ_{BRS} of P_{BRS} at $T = 10$ K. Γ_{BRS} was fitted from the low-frequency mode of the split P_{BRS} mode in the parallel channel. **g**, Thickness dependence of the fitted linewidth Γ_{BTS} of P_{BTS} at $T = 10$ K. The linewidth diverged when the thickness was decreased to the critical thickness ~10 nm. **h**, Thickness dependence of the fitted intensity ratio of I_{BTS}/I_{BRS} at $T = 10$ K. The intensity ratio I_{BTS}/I_{BRS}, defined as $(I_{BTS\text{-parallel}} + I_{BTS\text{-crossed}})/(I_{BRS\text{-l-parallel}} + I_{BRS\text{-l-crossed}})$, allowed us to focus on the relative strength between P_{BTS} and P_{BRS} and to exclude the possible impacts of the absolute intensity of P_{BTS} and P_{BRS} from different dielectric environments. The colour shading in **e**–**h** corresponds to regions II (yellow) and I (blue) and is the same as in Fig. 1a. t_c is the critical thickness for a crossover. **i**, Temperature dependence of the P_{BTS} mode for different thicknesses of $NiPS_3$. **j**, Temperature dependence of the P_{BRS} mode for different thicknesses of $NiPS_3$. Data points in **e**–**j** used the optimal fitting parameters with one standard error from fitting the Raman spectra. **k**, Thickness versus temperature phase diagram of $NiPS_3$.

the characteristic behaviour of the breathing mode). Figure 2d shows the temperature dependence of SW_{QES} for bilayer (2L), trilayer (3L), 4L, 13.1 nm and bulk $NiPS_3$. Each trace is normalized to its corresponding SW_{QES} at 200 K. There is a similarity among the SW_{QES} (T) plots, as the QES decreased as the temperature was reduced. Moreover, there was a small kink at a critical temperature around 155 K for bulk $NiPS_3$ ($T_{N,3D}$) and a broad bump around 120 K for 2L $NiPS_3$ ($T_{N,2D}$). More strikingly and more importantly, there was the contrasting behaviours of QES at low temperatures for $NiPS_3$ of distinct thicknesses. Bulk $NiPS_3$ had a fully suppressed QES at temperatures below a characteristic

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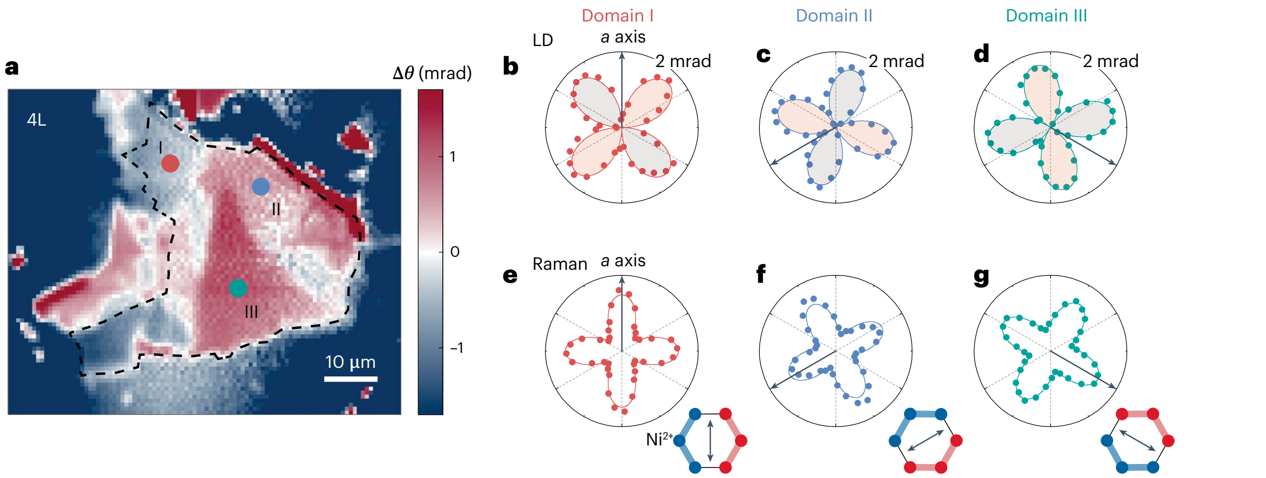

Fig. 4 | Polarization-dependent LD and Raman data for 4L $NiPS_3$. **a**, Spatially resolved LD map of 4L $NiPS_3$. The black dashed line outlines the boundary of the 4L $NiPS_3$ (optical image shown in Supplementary Fig. 15j). The three dots denote the three Potts-nematic domains with the nematic order parameter rotated 120° from one another. **b**–**d**, Angular-dependent polarization rotation $\Delta\theta$ in the nematic domains I (**b**), II (**c**) and III (**d**) in the 4L $NiPS_3$ sample shown in **a**. The red area has a positive sign and the grey area has a negative sign. The arrows indicate the *a* axis in the corresponding domain, which was determined by the polarization-dependent PL measurement (Supplementary Information Note 14). **e**–**g**, Polar plots of the polarization dependence of the fitted $P_{BRS\text{-}I}$ intensity at $T = 10$ K in the corresponding Potts-nematic domains I (**e**), II (**f**) and III (**g**) in the 4L $NiPS_3$ sample shown in **a**. The arrows denote the *a* axis in the corresponding domain. Insets, Schematics of the nematic order directors for the three different Potts-nematic domain states.

temperature $T_{sQES} \approx 90$ K, which is lower than $T_{N,3D}$. 13.1 nm $NiPS_3$ had a decreased T_{sQES} of ~40 K, whereas 2L, 3L and 4L $NiPS_3$ all had finite QES down to the lowest temperature 10 K. That is, T_{sQES}, when present, was lower than 10 K. Such a strong contrast in the spin fluctuations between the bulk and few-layer samples at lower temperatures is suggestive of different magnetic phases between three-dimensional (3D) and 2D $NiPS_3$.

It has already been convincingly shown that as the $NiPS_3$ thickness is reduce from 3D bulk to 2D flake, the spin fluctuations are clearly enhanced, which fulfils the key requirement for realizing vestigial nematicity in 2D $NiPS_3$. A natural next step was to examine the magnetic phases upon thinning 3D bulk $NiPS_3$ down to 2D films, using the right experimental tool to simultaneously detect BRS and BTS, both of which are required to distinguish zigzag AFM, Potts nematicity and paramagnetism, as illustrated in Fig. 1b–d. Thus, we used Raman spectroscopy. BRS can be detected through the splitting of doubly degenerate E_g phonons, and BTS can be probed by zone-folded phonon modes. Figure 3a shows Raman spectra for bulk $NiPS_3$ taken in linearly parallel and crossed channels at 10 K with a 633 nm excitation laser, over selected frequency ranges of interest (the full-range Raman spectra are in Supplementary Information Note 3). Figure 3b,c shows the temperature-dependent Raman spectra in the linearly parallel channel of Fig. 3a. First, the two Raman modes around 180 cm^{-1} are degenerate above $T_{N,3D} = 155$ K and split below $T_{N,3D}$. The frequency separation of these split modes has previously been assigned as the signature for zigzag AFM[34]. Despite the monoclinic stacking between honeycomb layers, the phonon modes obey the selection rules of D_{3d} above $T_{N,3D}$, and the two phonon modes at ~180 cm^{-1} belong to the $E_g(D_{3d})$ symmetry group. This case is BRS, rather than BTS, from the zigzag AFM order that lifts the $E_g(D_{3d})$ doublet into one $A_g(C_{2h})$ mode and one $B_g(C_{2h})$ mode (thus, it is referred to as P_{BRS}). As a result, the frequency separation was proportional to the composite nematic order parameter $\eta^\dagger\tau\eta$, instead of η. Second, the Raman mode at ~30 cm^{-1}, emerging below ~90 K, has not been reported previously when a 515 nm excitation laser was used[34,51]. Interestingly, this $T_{AFM,3D}$ = ~90 K onset temperature for the 30 cm^{-1} mode coincides with the temperature for the suppression of QES (T_{sQES}) and the temperature for the emergence of the zone-centre magnon at ~43 cm^{-1} (Supplementary Information Note 4). Because of this coincidence in the onset temperature and because of the following analysis, we consider this mode to be a phonon mode folded from the BZ boundary M point to the centre Γ point due to the zigzag AFM order. We first observed the in-plane and out-of-plane magnetic field independence for this mode and ruled out the possibility that it was a magnon[42,43] (Supplementary Information Note 5). We then compared to the calculated phonon dispersion spectra of bulk $NiPS_3$ and found no intralayer or interlayer Γ point phonons around 30 cm^{-1} (refs. 52,53). Considering that the zigzag AFM order has a wavevector $\mathbf{Q} = \mathbf{k}_M$, we checked the phonon spectra at the M point and, indeed, found a computed M point phonon mode at ~30 cm^{-1} (ref. 52). Furthermore, we examined the polarization dependence of this Raman mode and observed its clear anisotropy, confirming it to be a single-Q zone-folding process that breaks the threefold rotational symmetry (Supplementary Information Note 6). Note that the blueshift of this 30 cm^{-1} mode upon increasing the temperature possibly results from the thermal expansion of the lattice at higher temperatures (Supplementary Information Note 7). Thus, this zone-folded ~30 cm^{-1} mode is a direct consequence of BTS (labelled as P_{BTS}), and its strength should scale with the primary zigzag AFM order parameter η. Our results demonstrate that in bulk $NiPS_3$, the spins remain highly dynamic between 155 and ~90 K with BRS but no BTS ($T_{N,3D} > T > T_{AFM,3D}$). The long-range, static AFM order with both BRS and BTS forms below ~90 K ($T < T_{AFM,3D}$), which is consistent with the magnetic susceptibility measurements (Supplementary Information Note 8).

Having established the signatures for both BRS and BTS, as well as their relationship to the primary and composite order parameters, we proceeded to examine their evolution upon dimensionality reduction. Figure 3d shows the Raman spectra of P_{BRS} and P_{BTS} at 10 K for a range of thicknesses from bulk down to 2L. The signature of BTS, P_{BTS}, vanishes when the thickness was below ~10.6 nm, whereas the footprint for BRS, P_{BRS}, remained observable from bulk down to 2L $NiPS_3$. Although no observable BRS feature has been seen in 1L $NiPS_3$, the linewidth of the ~180 cm^{-1} E_g mode in 1L is ~10 cm^{-1}, which is much greater than both the linewidth and the splitting of P_{BRS} in 2L and thicker $NiPS_3$ (Supplementary Information Note 9). Moreover, the zone-centre magnon at ~43 cm^{-1} disappeared below ~10.6 nm, just like the BTS did (Supplementary Information Note 10). This contrast between the BTS and BRS features reveals that the zigzag AFM order in 3D bulk $NiPS_3$

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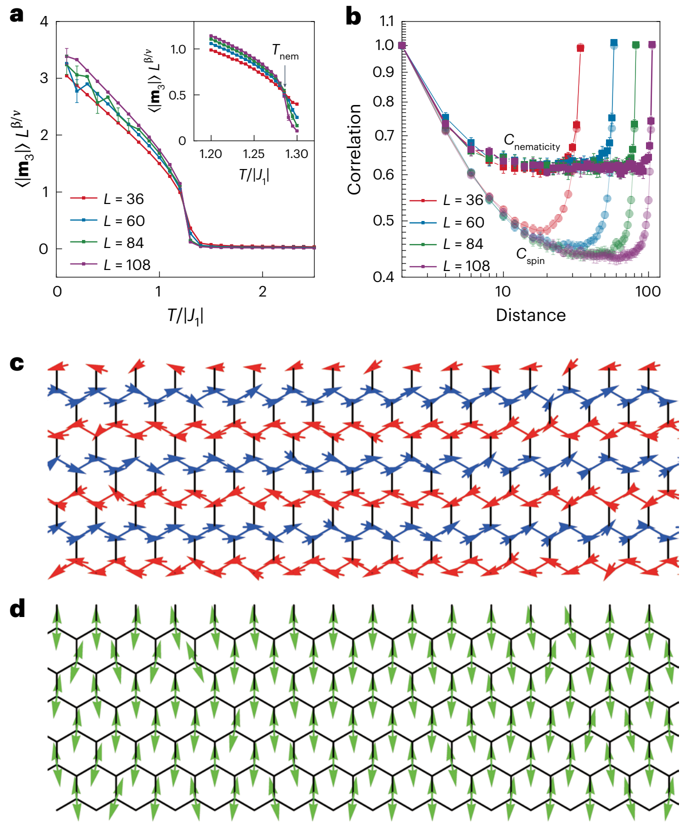

Fig. 5 | Monte Carlo calculations of the magnetic state in 2 L $NiPS_3$. **a**, Temperature dependence of the scaled Potts-nematic order parameter ($\langle|\mathbf{m}_3|\rangle$) for four different system sizes, $L = 36, 60, 84$ and 108 with $J_{l-n} = -0.07$ and $J_{l-nn} = 0$. Inset, enlargement around nematic phase transition temperature $T_{N,2L} = 1.285|J_1|$. The crossing point of the scaled nematic order parameter for different system size determines the nematic transition point. 2D Z_3 Potts exponents $\beta = 1/9$ and $\nu = 5/6$. **b**, Correlation of Potts nematicity ($C_{nematicity}$) and spin (C_{spin}) at temperature $T = 1.25|J_1| < T_{N,2L}$, as a function of distance for four different system sizes. The plot is in log–log scale. We normalized the two correlation functions at $r = 2$. Each data point in **a** and **b** is the average of 32 Monte Carlo bins with the same computational parameters and conditions. Each bin was measured with 10,000 Monte Carlo steps. The error bars are defined as their standard deviations. **c**, Snapshot of simulated spin configuration below $T_{N,2L}$, showing the lack of spin coherence. **d**, Image of simulated nematic director (normalized $\mathbf{m}_{3,i}$ configuration) for the spin state in **c**, showing the homogeneity of the nematicity.

transitions into the nematic order in 2D few-layer $NiPS_3$ across a critical thickness $t_C \approx 10.6$ nm. We ruled out the built-in strain in few-layer $NiPS_3$ as a possibility (Supplementary Information Note 11). Note that the survival of the BRS signature down to 2L is consistent with the presence of exciton photoluminescence (PL) down to 2L[31–33], as both are related to the nematic order parameter.

We further quantified the thickness dependence of both signatures with four important parameters: the frequency separation of P_{BRS} ($\Delta\omega_{BRS}$), the linewidth of P_{BRS} (Γ_{BRS}), the linewidth of P_{BTS} (Γ_{BTS}) and the intensity ratio between P_{BTS} and P_{BRS} (I_{BTS}/I_{BRS}). We discussed above how $\Delta\omega_{BRS}$ scales with the nematic order parameter. Its thickness independence in Fig. 3e confirms the robust presence of nematicity down to the bilayer. Furthermore, the nematicity coherence determines Γ_{BRS}, and the zigzag AFM coherence sets Γ_{BTS}. Figure 3f shows that Γ_{BRS} remained nearly constant until $t_C \approx 10.6$ nm and only started to increase slightly in 3L and more notably in 2L. In contrast, Fig. 3g illustrates the clear divergence of Γ_{BTS} on approaching t_C from above. The distinct trends between Γ_{BRS} and Γ_{BTS} across t_C clearly demonstrate the separation between the primary order (the zigzag AFM order of region I) and the vestigial order (the nematic order of region II) in the phase diagram illustrated in Fig. 1a. The suppression of I_{BTS}/I_{BRS} down to zero when thinning $NiPS_3$ across t_C (Fig. 3h) corroborates the melting of the zigzag AFM order but the survival of the vestigial nematicity below t_C. So that we could experimentally construct a thickness (dimensionality) versus temperature phase diagram for $NiPS_3$, we performed temperature-dependent measurements for selected thicknesses, including bulk, 74.0 nm, 25.4 nm, 13.1 nm and few-layer $NiPS_3$. Figure 3i,j shows the temperature dependence of the BTS and BRS features, respectively. The characteristic temperature for BTS (T_{AFM}) rapidly decreased, evolving from ~90 K for bulk to ~40 K for 13.1 nm $NiPS_3$, showing a notable consistency with T_{sQES}. In contrast, the temperature dependence of BRS has a clearly order-parameter-like onset (T_N) that remained nearly constant from bulk to 13.1 nm $NiPS_3$ (see Supplementary Information Note 12 for data for 2L–5L $NiPS_3$). We summarize the temperature and thickness dependence in the phase diagram in Fig. 3k, which shows regions I, II and III.

Having confirmed the intrinsic vestigial nematicity in few-layer $NiPS_3$ (thickness less than t_C) so that it is in region II of Fig. 1a, we next checked the symmetry class of this nematicity. We selected a large, 4L $NiPS_3$ sample (lateral dimension of ~40 μm) and imaged the nematic domains using scanning optical linear dichroism (LD) microscopy. The LD technique is a static probe known to be sensitive to BRS and has been used to study few-layer $NiPS_3$ (ref. 32). Although the BRS signature in the Raman processes around 180 cm^{-1} could potentially have been caused by a dynamic nematic liquid[54], the finite LD signal confirmed the static BRS of this nematic state in $NiPS_3$. For the same reason as for P_{BRS}, the LD signal scaled with the vestigial nematic order parameter, rather than the zigzag AFM order parameter, and therefore, the onset was at $T_{N,3D} \approx 155$ K for bulk $NiPS_3$ (Supplementary Information Note 13). Figure 4a shows a LD map for the 4L $NiPS_3$. Different domains produced different LD signals. There are clear boundaries between them. On surveying the angular dependence of the LD-induced polarization rotation across the 4L sample, we found that there were three, and only three, distinct patterns, as shown in Fig. 4b–d. These three patterns are related by the threefold rotational operation, corresponding to the three nematic domain states with the nematic order parameter rotated by 120° from one another that is, Z_3 Potts nematicity (schematic shown in the insets of Fig. 4e–g). The *a* axis direction, defined as the nematic order parameter orientation, was calibrated by the polarization-resolved PL measurements (Supplementary Information Note 14). These three Potts-nematic domain states were confirmed by the angular dependence of the lower frequency P_{BRS} mode (Fig. 4e–g), whose anisotropy originated from the reduction in the rotational symmetry due to the vestigial nematicity in the few-layer case. Thermal cycling did not change the nematic domains or the nematic order parameter orientations within domains (Supplementary Information Note 15), indicating that the nematic domain states were probably pinned by the structural monoclinic interlayer stacking[55]. Note that the weak coupling between the structural monoclinicity and the vestigial nematicity should not impact the physics of interest here, namely that enhanced fluctuations in 2D melt the zigzag AFM order and introduce the Z_3 Potts nematicity.

To comprehend the experimental finding of the crossover from 3D zigzag AFM to 2D Potts nematicity when the $NiPS_3$ thickness is decreased across t_C, we performed large-scale Monte Carlo simulations to visualize the magnetic phase for bilayer $NiPS_3$ and examine its primary (η) and composite ($\eta^{\dagger}\tau\eta$) order parameters. The magnetic state at finite temperature was simulated using a spin Hamiltonian with intralayer Heisenberg exchange coupling up to the third nearest neighbour (J_1, J_2 and J_3), strong easy-plane anisotropy (D_z), weak easy-axis anisotropy (D_x), and interlayer nearest and next-nearest exchange coupling (J_{l-n} and J_{l-nn}) (Methods). Note that the coordination for J_{l-n} obeys the threefold rotational symmetry whereas that for J_{l-nn} breaks the threefold rotational symmetry, and therefore, it encodes the monoclinic interlayer stacking. Figure 5a plots the temperature dependence of the nematicity order parameter $\eta^{\dagger}\tau\eta = \langle|\mathbf{m}_3|\rangle$ scaled with $L^{\beta/\nu}$ for four system sizes $L = 36, 60, 84$ and 108. The temperature is in units of $|J_1|$. β and ν are critical exponents for 2D Z_3 Potts nematicity. All four traces cross at a single temperature (inset of Fig. 5a), which defines the nematicity critical temperature, $T_{N,2L} = 1.285\,|J_1| = 67.1$ K. This is of the same order as the experimental value of 120 K in Fig. 2d. This observation clearly confirms the emergence of Potts nematicity in 2L $NiPS_3$ (see Supplementary Information Note 16 for consistent results for 1L to 5L $NiPS_3$). Note that such a 2D Potts-nematic phase transition remains when the effect from the monoclinic stacking (J_{l-nn}/J_{l-n}) is small, as with our experimental findings. However, it can evolve into a crossover when J_{l-nn}/J_{l-n} becomes larger (Supplementary Information Note 17). We contrast this 2D Potts-nematic state against the zigzag AFM order by computing and comparing the nematicity and spin correlation, $C_{\text{nematicity}}(r)$ and $C_{\text{spin}}(r)$, at a temperature $T = 1.25|J_1| < T_{N,2L}$, as shown in Fig. 5b. After the initial decay over a short distance for both curves, which are normalized to their corresponding values at $r = 2$, $C_{\text{nematicity}}(r)$ reaches a saturation plateau that is consistent across all four system sizes. By contrast, $C_{\text{spin}}(r)$ decays further s the system size increases. Note that the increase at distances greater than half of the system size is due to the finite-size effect. The size-independent plateau behaviour for $C_{\text{nematicity}}(r)$ confirms the long-range order of the composite nematic order parameter, but the size-dependent ever-decaying trend for $C_{\text{spin}}(r)$ demonstrates the 'disordered' nature of the zigzag AFM state, which corroborates our experimental findings and shows that the state belongs to region II in Fig. 1a. To directly visualize the real-space spin arrangements for this unique magnetic phase, we plot a snapshot of the simulated result in Fig. 5c for spin texture and in Fig. 5d for the normalized nematicity director texture. The spins (red and blue arrows) in Fig. 5c are randomly canted away from the zigzag chain direction, thus losing the spin coherence at long distance, as revealed by Fig. 5b. As a comparison, the nematicity directors (green double arrows) in Fig. 5d show a much better homogeneity and therefore there is long-range coherence, as found in Fig. 5b.

Our work on the thickness dependence of spin fluctuations and magnetic phases in $NiPS_3$ clearly demonstrates that the enhanced fluctuations in 2D introduce the Z_3 Potts-nematic state by partially melting the conventional zigzag AFM order. Based on this result, we envision several emergent research opportunities. First, it may be promising to exploit the enhanced fluctuations in 2D, rather than suppressing them, to discover spin-fluctuation-driven magnetic phases, including and going beyond this Z_3 spin-induced Potts nematicity. This is a different direction to present mainstream efforts in 2D magnetism research. Second, it remains an open experimental question what the spin coherence length is in 2D $NiPS_3$, whereas the corresponding coherence length of the Potts nematicity is much beyond optical wavelengths (~1 μm). High-spatial-resolution experimental techniques (approximately nanoscale) are required to solve this problem. Third, because the monoclinic interlayer stacking provides an effective field for coupling with the Potts nematicity in 2D $NiPS_3$, it would be interesting to explore what the magnetic state would be if two $NiPS_3$ flakes were twisted to create a moiré superlattice whose supercell contains all three monoclinic stacking geometries. Finally, given that 3D $NiPS_3$ realizes the Mott insulating state with strong electron correlations[56] and also hosts ultra-narrow exciton emission lines, possibly due to the many-body effect[31,57], 2D $NiPS_3$ offers a unique platform for exploring the interplay between the spin degree of freedom with strong fluctuations and the charge degree of freedom with strong correlations, a parameter regime previously inaccessible in 3D systems.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41567-024-02618-6.

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Methods

Sample fabrication

High-quality $NiPS_3$ single crystals were grown by chemical vapour transport. Few-layer $NiPS_3$ samples were prepared on Si/SiO_2 by mechanical exfoliation from high-quality bulk $NiPS_3$ single crystals in a nitrogen-filled glovebox with oxygen level below 0.1 ppm and water level below 0.5 ppm. To measure the NV magnetometry, one hexagonal boron nitride (hBN) thin flake was first picked up with a poly(bisphenol A carbonate) stamp. Then, selected few-layer $NiPS_3$ samples on Si/SiO_2 were picked up by the hBN flake on the poly(bisphenol A carbonate) stamp. The hBN/few-layer $NiPS_3$ was transferred onto a diamond membrane for the NV magnetometry measurements. The thicknesses of the few-layer samples were first determined based on their optical contrast, which was confirmed by atomic force microscopy after the measurements.

Raman spectroscopy

Raman spectroscopy measurements were performed with $\lambda = 532$ nm and 633 nm excitation lasers. The laser beam was focused onto the sample down to ~2–3 µm in diameter. The laser power was kept below 1 mW to minimize local heating during measurements. The backscattering geometry was used. The scattered light was detected by a thermoelectrically cooled charge-coupled device (CCD) camera from Horiba Scientific, which had a spectral resolution of 0.4 cm^{-1}. The temperature-dependent Raman spectra were measured in a commercial variable-temperature, closed-cycle cryostat (Cryo Industries of America). For the magnetic-field-dependent Raman spectroscopy, a commercial superconducting magnet (Cryo Industries of America) was used to achieve a magnetic field from 0 to 7 T.

Fitting the Raman spectra

We used several Lorentzian functions, $\sum_N \frac{A_N(\Gamma_N/2)^2}{(\omega-\omega_N)^2+(\Gamma_N/2)^2} + C$, to fit the P_{BTS} and P_{BRS} modes, where A_N is the peak intensity, ω_N is the peak position, Γ_N is the peak linewidth and C is the background with $N = 1$ for a single peak and $N = 2$ for double peaks. For the QES signal, we first removed the breathing mode signal if present and then fitted the Raman data in the Raman shift range 8–40 cm^{-1} using the Lorentzian function centred at 0 cm^{-1}, $\frac{A_N(\Gamma_N/2)^2}{\omega^2+(\Gamma_N/2)^2} + C$, where C is a constant for the background. Finally, we integrated the fitted curves without a background in the range 8–40 cm^{-1} as the QES intensity.

Derivation of angular dependence of Raman modes

We derived the polarization dependence of the Raman intensity in the parallel channel as a function of the polarization rotation angle θ. The measured Raman intensity (I) is proportional to the square of the product of incident light polarization ($\hat{e}_i$), Raman tensor (R) and the scattered light polarization ($\hat{e}_f$): $I \propto |\langle \hat{e}_i|R|\hat{e}_f\rangle|^2$. In the parallel channel, the incident light

$$|\hat{e}_i\rangle = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix},$$

and the scattered light

$$|\hat{e}_f\rangle = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}.$$

The Raman tensor for the P_{BTS} mode is $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$. So, the intensity

$$I_{BTS} \propto \left|(\cos\theta, \sin\theta)\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}\begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}\right|^2 = \frac{1}{8}(a-b)^2\cos 4\theta$$
$$+\frac{1}{2}(a^2-b^2)\cos 2\theta + \frac{1}{8}(3a^2+2ab+3b^2)$$

(Supplementary Fig. 6a). The Raman tensors for $P_{BRS\text{-}l}$ and $P_{BRS\text{-}h}$ are $\begin{pmatrix} e & 0 \\ 0 & -e \end{pmatrix}$ and $\begin{pmatrix} 0 & f \\ f & 0 \end{pmatrix}$, respectively. So the intensities

$$I_{BRS-l} \propto \left|(\cos\theta, \sin\theta)\begin{pmatrix} e & 0 \\ 0 & -e \end{pmatrix}\begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}\right|^2 = \frac{e^2}{2}(\cos 4\theta + 1)$$

and

$$I_{BRS-h} \propto \left|(\cos\theta, \sin\theta)\begin{pmatrix} 0 & f \\ f & 0 \end{pmatrix}\begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}\right|^2$$
$$= \frac{f^2}{2}(1-\cos 4\theta) = \frac{f^2}{2}\left(\cos 4\left(\theta + \frac{\pi}{4}\right) + 1\right).$$

Patterns for both $P_{BRS\text{-}l}$ and $P_{BRS\text{-}h}$ show fourfold symmetry and have a 45° phase shift from one another (Supplementary Fig. 6b).

NV spin relaxometry measurement

The wide-field NV spin relaxation rate was measured in a closed-cycle optical cryostat with the measurement temperature ranging from 4.5 to 350 K. A microsecond-long green laser pulse generated by an electrically driven 515 nm laser was used to initialize the NV centres to the $m_s = 0$ state and read out the spin-state-dependent PL. The laser beam spot focused on the diamond surface was around 20 µm × 20 µm. A CMOS camera was used to collect the NV fluorescence images. The pulses used to trigger the camera exposures and to drive the laser were controlled by a programmable pulse generator. Nanosecond-long microwave pulses were generated by sending continuous microwave currents to a microwave switch electrically controlled by a programmable pulse generator. These were then delivered to an Au strip line patterned onto diamond samples. An external field of ~70 G was generated by a cylindrical NdFeB permanent magnet attached to a scanning stage inside the optical cryostat. Further details of the NV relaxometry measurement protocol can be found in Supplementary Information Note 1.

LD measurement

The LD was measured with a laser diode at wavelength 635 nm (Thorlabs PL202) and 20 µW power. A photo-elastic modulator (PEM; PEM-100, Hinds Instruments) on the incident path was used to measure the LD. The incident light with linear polarization at angle 45° with respect to the PEM fast axis was modulated by a mechanical chopper and the PEM with a retardance of $\lambda/2$ by sequence. The light was then passed through a half-wave plate and focused onto the sample at normal incidence with a ×50 objective lens. The reflected light passed through the half-wave plate again and then went into the detector through a plate beam splitter. The reflected light was measured by a balanced amplified photodetector (ThorLabs PDB210A). The signal was demodulated by two lock-in amplifiers (Zurich Instruments, MFLI 500 kHz/5 MHz) separately, of which one lock-in amplifier was referenced to the second harmonic of the PEM frequency $2f = 84.183$ kHz and the other lock-in amplifier was referenced to the chopping frequency $f = 511$ Hz to get the total reflectance as a normalization. For the polarization-dependent LD measurement, the polarization direction of the incident light was rotated by the half-wave plate. Thus, the polarization rotation $\Delta\theta$ induced by the LD of few-layer $NiPS_3$ can be written as $\Delta\theta = \mathrm{LD}\sin(2\alpha + \varphi)$, in which α is the polarization direction of the incident light and the phase φ depends on the relative angle between the crystal axis and the polarization direction of the incident light at zero. $\Delta\theta$ is zero when the polarization direction of the incident light is along the direction of the crystal axes. The angle-dependent signal was fitted by the equation $\Delta\theta = \mathrm{LD}\sin(2\alpha + \varphi)$. A small non-zero background from the inevitable birefringence of the optical components in the whole set-up was subtracted during the fitting process. For the LD scanning measurement, a two-axis Galvo scanning mirror paired with a confocal imaging system was used to move the light spot on the sample in normal incidence with a constant fluence. The half-wave plate was fixed at a certain angle during scanning.

Polarization-dependent PL measurement

The PL was measured with a continuous-wave solid-state laser at 532 nm (Thorlabs) to excite the few-layer $NiPS_3$ sample. The beam focused on the sample was ~2.5 µm in diameter. The collected signals were detected by a spectrometer (Princeton Instruments) with a CCD camera. The sample was kept at 5 K using a Montana CA100 system. For the polarization-dependent PL measurements, a polarizer was put in the detection beamline to change the polarization direction of the light collected by the CCD camera.

Monte Carlo simulations

The magnetic Hamiltonian of $NiPS_3$ can be described by a 2D XY-type spin Hamiltonian with easy-plane single-ion anisotropy ($D_z > 0$) and easy-axis single-ion anisotropy ($D_x < 0$)[43]:

$$H = \sum_{\langle ij \rangle} J_{ij}\left(S_i^x S_j^x + S_i^y S_j^y + S_i^z S_j^z\right) + D_z \sum_i \left(S_i^z\right)^2 + D_x \sum_i \left(S_i^x\right)^2,$$

in which J_{ij} are the exchange coupling parameters, D_z is the easy-plane single-ion anisotropy and D_x is the easy-axis single-ion anisotropy. We considered the nearest exchange coupling interaction J_1, the next-nearest exchange coupling interaction J_2, the third-nearest exchange coupling interaction J_3 and the nearest exchange coupling interaction between neighbouring layers $J_{\text{l-n}}$. We performed large-scale Monte Carlo simulations on bilayer $NiPS_3$ systems. In our simulations, we employed various techniques including local updates, Woff updates and the over-relaxation technique. The spins were treated as classical O(3) vectors, and the simulation was warmed up with 10^4 Monte Carlo steps. In our simulation, we set the parameters $\{J_1, J_2, J_3, J_{l-n}, D_x, D_z\}$ to be $\{-1, 0, 2.22, -0.07, -0.002, 0.047\}$, with $|J_1|$ as the unit[35,43]. We simulated the bilayer case ($L_z = 2$) with periodic boundary conditions along the x and y directions but with open boundary conditions along the z direction. The linear system ranged in size from $L = 36$ to 108. Note that the the opposite signs of J_1 and J_3 introduced geometric frustrations, which can cause a sign problem in quantum Monte Carlo simulations. Therefore, we used classical Monte Carlo simulations with $S = 1$ for this current study. Our simulated results captured well the nematic phase transition of this 2D XY magnetic system, which supports the validity and sufficiency of the classical Monte Carlo simulations used here.

We checked the spin Hamiltonian with nearest- and next-nearest-neighbour interlayer exchange coupling for few-layer $NiPS_3$, where

$$\begin{aligned} H = &\sum_{\langle i,j \rangle} J_1 \mathbf{S_i} \cdot \mathbf{S_j} + \sum_{\langle\langle i,j \rangle\rangle} J_2 \mathbf{S_i} \cdot \mathbf{S_j} + \sum_{\langle\langle\langle i,j \rangle\rangle\rangle} J_3 \mathbf{S_i} \cdot \mathbf{S_j} \\ &+ \sum_{\{i,j\}} J_{l-n} \mathbf{S_i} \cdot \mathbf{S_j} + \sum_{\{\{i,j\}\}} J_{l-nn} \mathbf{S_i} \cdot \mathbf{S_j} + \sum_i D_z \left(S_i^z\right)^2 + \sum_i D_x \left(S_i^x\right)^2. \end{aligned}$$

J_{l-n} and J_{l-nn} are the nearest- and next-nearest-neighbour interlayer exchange coupling, respectively. The bond environment of J_{l-nn}, $\{\{i,j\}\}$ accounts for the broken threefold rotational symmetry from the monoclinic stacking.

In our simulation, we define the order parameter of the Z_3 rotational symmetry $\mathbf{m}_3$ as[27]

$$\mathbf{m}_3 = \boldsymbol{\sigma}_1 \epsilon_1 + \boldsymbol{\sigma}_2 \epsilon_2 + \boldsymbol{\sigma}_3 \epsilon_3.$$

Here, $\boldsymbol{\sigma}_i$ represents the bond direction: $\boldsymbol{\sigma}_1 = (0, 1)$, $\boldsymbol{\sigma}_2 = (-\sqrt{3}/2, -1/2)$ and $\boldsymbol{\sigma}_3 = (\sqrt{3}/2, -1/2)$. ϵ_i is the average neighbouring bond energy along the $\boldsymbol{\sigma}_i$ direction: $\epsilon_i = \mathbf{S_r} \cdot \mathbf{S_{r+e_i}}$. This definition of the nematic order parameter is consistent with $\eta^\dagger \tau \eta = \left(\frac{\sqrt{3}}{2}(\eta_3^2 - \eta_2^2), \frac{1}{2}\left(2\eta_1^2 - \eta_2^2 - \eta_3^2\right)\right)$ by writing $\eta^\dagger \tau \eta$ as $\eta_1^2 \boldsymbol{\sigma}_1 + \eta_2^2 \boldsymbol{\sigma}_2 + \eta_3^2 \boldsymbol{\sigma}_3$ where $\boldsymbol{\sigma}_j = \left(\cos\left[\frac{2\pi}{3}(j-1) + \frac{\pi}{2}\right], \sin\left[\frac{2\pi}{3}(j-1) + \frac{\pi}{2}\right]\right)$. Noting that $\eta_j^2 = \eta_j \cdot \eta_j = \left(\mathbf{S_r} - \mathbf{S_{r+e_j}}\right) \cdot \left(\mathbf{S_r} - \mathbf{S_{r+e_j}}\right) = -2\mathbf{S_r} \cdot \mathbf{S_{r+e_j}} + \mathbf{S_r^2} + \mathbf{S^2_{r+e_j}}$, where the last two terms $\mathbf{S_r^2}$ and $\mathbf{S^2_{r+e_j}}$ are constants, $\eta_1^2 \boldsymbol{\sigma}_1 + \eta_2^2 \boldsymbol{\sigma}_2 + \eta_3^2 \boldsymbol{\sigma}_3 = -2\sum_j \left(\mathbf{S_r} \cdot \mathbf{S_{r+e_j}}\right) \boldsymbol{\sigma}_j + \text{const} \sum_j \boldsymbol{\sigma}_j = -2\sum_j \left(\mathbf{S_r} \cdot \mathbf{S_{r+e_j}}\right) \boldsymbol{\sigma}_j$ because $\sum_j \boldsymbol{\sigma}_j = 0$. Therefore, $\eta^\dagger \tau \eta = \left(\frac{\sqrt{3}}{2}(\eta_3^2 - \eta_2^2), \frac{1}{2}\left(2\eta_1^2 - \eta_2^2 - \eta_3^2\right)\right)$ and $\mathbf{m}_3$ differ from each other only by a constant factor of 2.

The norm of $\mathbf{m}_3$, $\langle|\mathbf{m}_3|\rangle$, is the nematic order parameter. If Z_3 rotation symmetry is present, $\langle|\mathbf{m}_3|\rangle$ will vanish. However, $\langle|\mathbf{m}_3|\rangle$ will become non-zero under Z_3 rotational symmetry breaking. Moreover, we measured the nematic correlation, $C_{m_3}(r) = \langle \mathbf{m}_{3,i} \mathbf{m}_{3,i+r} \rangle$, in which the nematic director $\mathbf{m}_{3,i} = \boldsymbol{\sigma}_1 \epsilon_{i,1} + \boldsymbol{\sigma}_2 \epsilon_{i,2} + \boldsymbol{\sigma}_3 \epsilon_{i,3}$ is defined on each lattice site i. By examining the behaviour of $\langle|\mathbf{m}_3|\rangle$ and $C_{m_3}(r)$, we can identify the occurrence of nematic order. We also measured the spin correlation, $C_s(r) = \langle \mathbf{S}_i \mathbf{S}_{i+r} \rangle$, to investigate whether long-range spin order occurs. We also performed the calculations with the binder ratio to compare with our temperature dependence of the nematic order parameter (Supplementary Information Note 17).

Phonon calculations

The first-principles simulations were performed using the Vienna ab initio Simulation Package (VASP). The atomic structure was optimized within the generalized gradient approximation and the Perdew–Burke–Ernzerhof exchange-correlation functional. The Monkhorst–Pack k-mesh was set to $7 \times 7 \times 1$ with a 550 eV kinetic cutoff energy. All structures were fully relaxed until the force on each atom was less than 0.02 eV $Å^{-1}$. To avoid spurious interactions between periodic images, the vacuum space was set to be more than 15 Å. Phonon calculations were performed by the supercell approach, wherein supercells containing $2 \times 2 \times 1$ unit cells were used. The real-space force constants were calculated in the density-functional perturbation theory implemented in the VASP code, and phonon frequencies were calculated from the force constants using the PHONOPY code.

Data availability

Source data are provided with this paper. All other data that support this study are available from the corresponding authors upon reasonable request.

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Author contributions

Z.S., R.H. and L.Z. conceived the idea and initiated this project. Z.S. exfoliated the $NiPS_3$ thin flakes with different layer numbers. G.Y., Z.Y., C.N. and Z.S. carried out the Raman experiments under the supervision of L.Z. and R.H. M.H. performed the NV spin relaxometry under the supervision of C.D. C.Z. carried out the Monte Carlo simulations under the supervision of K.S. and Z.Y.M. Q.L. and Z.S. carried out the atomic force microscopy measurements and the PL measurements guided by H.D. and L.Z. N.H. grew the high-quality $NiPS_3$ bulk single crystals under the supervision of D.M. G.Z. performed the susceptibility measurements under the supervision of L.L. X.X. performed the phonon calculations under the supervision of L.Y. Z.S., R.H. and L.Z. analysed the data and wrote the manuscript. All authors participated in discussions about the results.

Competing interests

The authors declare no competing interests.

Additional information

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